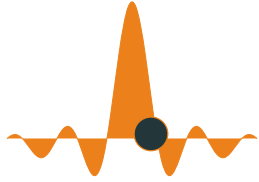
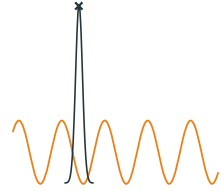
Spontaneous collapse models are Bohmian mechanics applied to a hidden heat bath

Antoine Tilloy, with Howard M. Wiseman
Max Planck Institute of Quantum Optics, Garching, Germany





Workshop: New Topics in Quantum Foundations
Université de Lausanne, Switzerland
November 30th, 2018

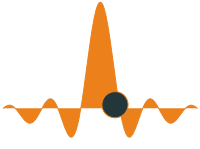


Alexander von Humboldt
Stiftung/Foundation

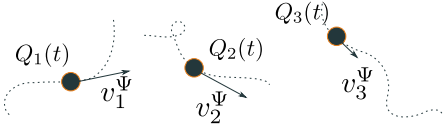


Introduction: two realist reconstructions of QM

Bohm



- *"Particles move"*

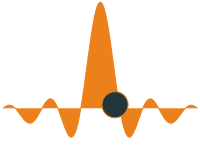


- The wavefunction evolves unitarily

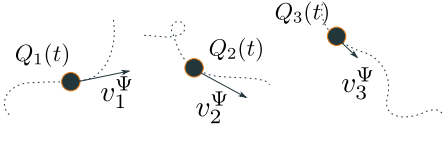
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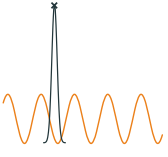
♠ *"Particles move"*



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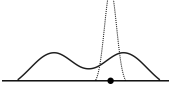
Collapse



♠ The wavefunction does **not** evolve unitarily

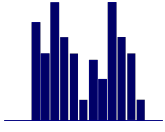
$$\partial_t|\psi\rangle = -iH|\psi\rangle + \varepsilon\, f(\psi, w)\,|\psi\rangle$$

♠ Beables are constructed from w or ψ



Introduction: differences

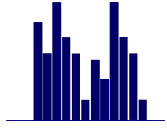
1. **Empirically different**



→ collapse model predictions deviate from those of the **Standard Model**

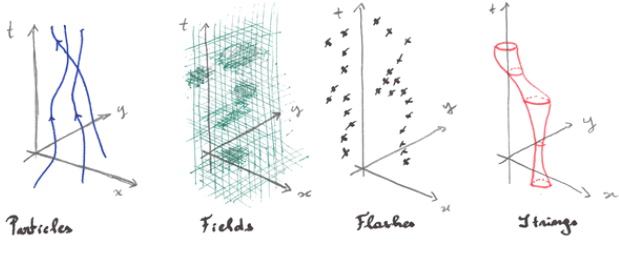
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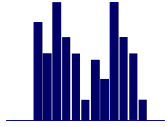
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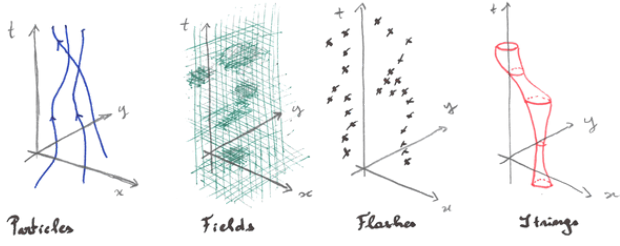
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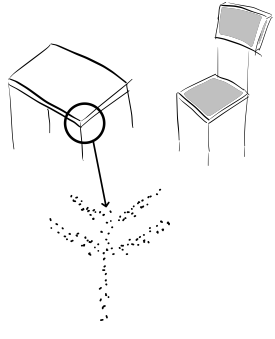
2. **Ontologically different**



3. **Different spirit**
   "Modify the Schrödinger equation" *vs* "Add hidden variables"

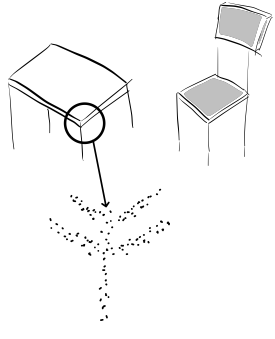
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2. Dynamical reduction works in similar ways – [Toros et al. 2016]

$$\psi_C^{\text{Bohm}}(x_{\text{bath}}, \cdot) \sim \psi^{\text{GRW}}(\cdot)$$

Equivalence

Weak Equivalence

For a given collapse model on a system S, there exists a bath B such that Bohmian mechanics on $\{S + B\}$ has the same empirical content.

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This is **true**, often **overlooked** but rather **trivial**:

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$$\exists \;\; H \;\; \text{and} \;\; |\Psi_0\rangle = |\psi_0\rangle \otimes |\text{aux}\rangle \;\; \text{s.t.} \;\; \partial_t |\Psi_t\rangle = -iH|\Psi_t\rangle \;\; \text{and} \;\; \text{tr}_B\Big[\, |\Psi_t\rangle\langle\Psi_t| \,\Big] = \rho_t$$

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4. This dilated unitary dynamics can be Bohmianized

Equivalence

Strong Equivalence

For a given collapse model on a system S, there exists a bath B such that Bohmian mechanics on $\{S + B\}$ contains the exact same objects:

- The Bohmian conditional wave function **is** the collapse model stochastic wave function:

$$|\psi^{\text{collapse}}\rangle = |\psi_C^{\text{Bohm}}(x_{\text{bath}})\rangle$$

- The Bohmian particle positions are in one to one correspondence with the collapse noise

$$x_{\text{bath}}^{\text{Bohm}} \underset{\textit{Fourier}}{\longleftrightarrow} \textbf{collapse noise } w$$

This result is much less trivial

Precise formulation

Collapse model

For states $|\psi\rangle$ in a Hilbert space $\mathscr{H}_s$, we are given a collapse model "collapsing" a set of operators A_i, $i = 1 \cdots d$, with a colored noise field $w_i(t)$:

$$\partial_t |\psi_w\rangle = \underbrace{-iH|\psi\rangle}_{\text{linear QM}} + \underbrace{f(A_i, w_i, \psi_w)\,|\psi_w\rangle}_{\text{collapse term}}$$

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Bohmian mechanics on extend space

One can find a bath Hilbert space $\mathscr{H}_b$ made of harmonic oscillators $\{x_{i,\omega}\}$ indexed by $i = 1 \cdots d$ and a frequency $\omega \in \mathbb{R}$, and an interaction Hamitonian H_{int} such that:

1. $|\Psi(0)\rangle = |\psi(0)\rangle \otimes |0\rangle \in \mathscr{H}_{\text{total}} = \mathscr{H}_s \otimes \mathscr{H}_b$
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Comment: the interaction Hamiltonian is not crazily fine tuned:
$H_{\text{int}} = \sum_{i=1}^{d} A_i \otimes \int_{\mathbb{R}} d\omega\, \kappa^{\ell}_{i,\omega}\, p_{\ell,\omega}$ where $\kappa^{\ell}_{i,\omega}$ is the square root of the **Fourier transform** of the noise field correlation function

Comments

1. Technically, the proof uses an "old" result $\rightarrow$ `[Gambetta & Wiseman 2003]`
2. The result is **valid** for continuous collapse models with **non-white noise**
3. The result is still valid for continuous collapse models with white noise, but unimpressive
4. To obtain the result, one must first correct mistakes in the collapse literature...
5. The result is obtained in a bruteforce way, with an explicit (difficult) computation

General continuous collapse models

One starts with a **non-linear** stochastic Schrödinger equation:

$$\partial_t |\psi\rangle = -iH|\psi\rangle + \varepsilon \underbrace{f(\psi, w)}_{\text{non-linear}} |\psi\rangle \tag{1}$$

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- Requiring that equation (1) reduces superpositions of certain operators (position related).
- Imposing a **linear** master equation for $\rho_t = \mathbb{E}\Big[\, |\psi_t\rangle\langle\psi_t| \,\Big]$:

$$\rho_t = \Phi_t \cdot \rho_0 \quad CPTP$$

Required for the probabilistic interpretation (Born rule, no faster-than-light signalling).

Collapse models with white noise

For white noise, one can find a non-linear stochastic differential equation directly:

$$\mathrm{d}|\psi(t)\rangle = \Big[-iH\,\mathrm{d}t + \sqrt{\gamma}\,(A_i - \underbrace{\langle A_i\rangle_t}_{\text{non-linearity}})\,\mathrm{d}\underbrace{W_i(t)}_{\text{noise}} - \frac{\gamma}{2}(A_i - \underbrace{\langle A_i\rangle_t}_{\text{non-linearity}})^2\,\mathrm{d}t\Big]|\psi(t)\rangle$$

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But... **white noise** is weird and soon falsified $\rightarrow$ go **colored**

Collapse models with colored noise: linear evolution

Linear differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}|\phi_w(t)\rangle = \Big[-iH + \underbrace{\sqrt{\gamma}\, w_i(t) A_i}_{\text{noise drive}} - \underbrace{2\sqrt{\gamma}\, A_i \int_0^t \mathrm{d}s\, D_{ij}(t,s) \frac{\delta}{\delta w_j(s)}}_{\text{weird necessary memory term}} \Big] |\phi_w(t)\rangle,$$

with **colored** noise field:

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Reverse construction: done the other way around in the open quantum system literature!
unitary evolution $\overset{\text{trace out}}{\longrightarrow}$ **open evolution** $\overset{\text{“unraveling”}}{\longrightarrow}$ **stochastic Schrödinger equation**

Collapse models with colored noise: normalization

Normalization

$$|\psi_w(t)\rangle = \frac{1}{\sqrt{\langle\phi_w(t)|\phi_w(t)\rangle}}|\phi_w(t)\rangle$$

but now $|\psi_w(t)\rangle$ does not yield a linear master equation.

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Naive probability cooking

Define a new probability $\mathrm{d}\mathbb{P}(t) := \langle\phi_w(t)|\phi_w(t)\rangle\,\mathrm{d}\mathbb{Q}$ such the average ρ_t is not changed by the normalization:

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Problem: $\mathbb{P}(t_1)[w]$ and $\mathbb{P}(t_2)[w]$ disagree even restricted to the common past of t_1 and t_2.

Collapse models with colored noise: correct normalization

Instead of changing the probability with time, change the noise field:

1. Take a new noise trajectory $w^{[t]} : u \to w^{[t]}(u)$ at each time t such that for all functionals f:

$$\mathbb{E}_{\mathbb{Q}}\Big[f(w^{[t]})\Big] = \mathbb{E}_{\mathbb{P}(t)}\Big[f(w)\Big].$$

   [Strunz, Gisin, Diosi 1999] [Wiseman, Gambetta 2002]

2. Take the state $t \to |\psi_{w^{[t]}}(t)\rangle$ as output where $|\psi_w(t)\rangle = \frac{1}{\sqrt{\langle\phi_w(t)|\phi_w(t)\rangle}}|\phi_w(t)\rangle$

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Explicit noise expression

Horrible computations yield:

$$w_j^{[t]}(s) = w_j(s) + 2\sqrt{\gamma}\int_0^t \mathrm{d}\tau\, D_{ij}(\tau, s)\langle A_i\rangle_\tau,$$

where:

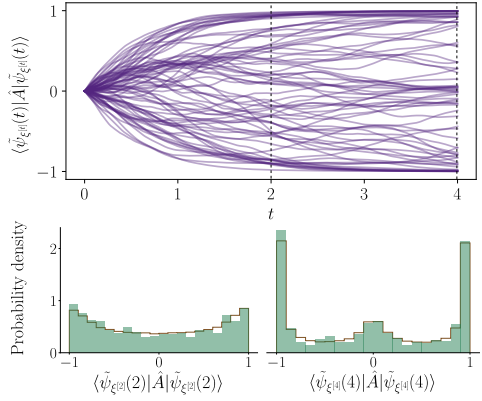
$$\langle A_i\rangle_\tau = \frac{\langle\phi_{w^{[\tau]}}(\tau)|A_i|\phi_{w^{[\tau]}}(\tau)\rangle}{\langle\phi_{w^{[\tau]}}(t)|\phi_{w^{[\tau]}}(\tau)\rangle} = \langle\psi_{w^{[\tau]}}(\tau)|A_i|\psi_{w^{[\tau]}}(\tau)\rangle.$$

Comments on collapse models with colored noises

- A collapse model with colored noise is not just equivalent with putting noise in the Schrödinger equation
- Preserving linearity of the master equation requires a complicated non-Markovian term: **the wave function derivative depends on the whole past of the noise field**
- Preserving the norm requires a complicated continuous noise redefinition: **while the state evolves, the whole noise trajectory evolves**
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But still... well defined, computable

Bohmian mechanics in a general context

Guiding law

$$\frac{\mathrm{d}}{\mathrm{d}t} x_k(t) = v_k(x(t), |\psi_t\rangle).$$

Velocity field

$$v_k(x, |\psi_t\rangle) = \frac{\mathfrak{Re}\left[\langle\psi_t|x\rangle\langle x|\hat{V}_k|\psi_t\rangle\right]}{\langle\psi_t|x\rangle\langle x|\psi_t\rangle}$$

Velocity operator

$$\hat{V}_k = -i\left[\hat{X}_k, \hat{H}\right].$$

Sketch of the proof

Bath Hilbert space $\mathscr{H}_b$ made of harmonic oscillators $\{x_{i,\omega}\}$ indexed by $i = 1 \cdots d$ and a frequency $\omega \in \mathbb{R}$, and an interaction Hamitonian $H_{\text{int}} = \sum_{i=1}^{d} A_i \otimes \int_{\mathbb{R}} \mathrm{d}\omega \, \kappa_{i,\omega}^{\ell} \, p_{\ell,\omega}$

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   [`Bohm side`] – **Conditional** wave function $|\psi_x^C(t)\rangle$ for **fixed** bath particle positions

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2. [`Collapse side`] – noise field w
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$$w_i(x,s) = \sqrt{2} \int_{\mathbb{R}^+} \mathrm{d}\omega\, \kappa_{i,\omega}^{\ell} \left(\cos(\omega s) x_{\ell,\omega}^{+}(t) + \sin(\omega s) x_{\ell,\omega}^{-}(t) \right)$$

choosing $x_{i,\omega}$ distributed according to the Born rule gives the correct distribution for w_i

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4. Putting all together $|\psi_w(t)\rangle = |\psi_{x(t)}^C(t)\rangle$

Lessons & questions

1. We should be careful of distinguishing **quantum theory** from its instantiation in the **Standard Model**
2. The two leading approaches to down to earth foundations of quantum theory are formally the same: **not** a duality, **not** even just empirically equivalent, ontologies in one to one
3. The distinction between a **deterministic** and a **stochastic** theory is inevitably blurry
   - In collapse models randomness is progressively revealed
   - In Bohmian mechanics it is in the initial condition

   Same modulo a Fourier transform...
4. Is there another way to construct realist interpretations without **guiding laws**? Kent? My attempt with "all at once random fields"?